آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-I Biology Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of Secondary School Certificate (HSSC) Part I Biology. It contains comments on candidates' responses to the 2025 HSSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. It is imperative to refer to command word guide available on AKU-EB website for understanding the expectations of the command words.

General Observations

Most candidates performed commendably in their responses, particularly in explaining the mechanisms of gaseous exchange in fish, the structural adaptations of mitochondria and chloroplasts, and the distinguishing characteristics of the different classes of the phylum Annelida. Nonetheless, it is imperative that teachers place greater emphasis on the following areas and provide targeted exercises and supplementary practice to foster a more comprehensive grasp of the subject matter:

- The structural differences between types of fatty acids such as acylglycerol (triglyceride), a phospholipid and a terpene.
- The construction and interpretation of graphs related to different concepts of biology such as enzyme activity etc. Real-life examples of plant and animal cells with reference to different cellular organelles.
- Reasoning related to the classification of major groups of tracheophytes.

- The identification of companion cells in the diagrams, along with their structure and functions.
- The effects of smoking on the human respiratory system.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

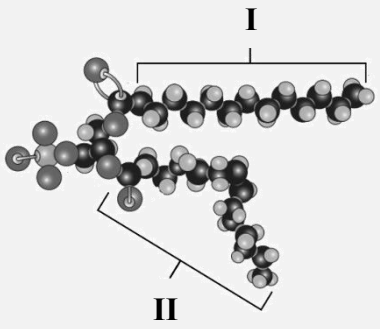
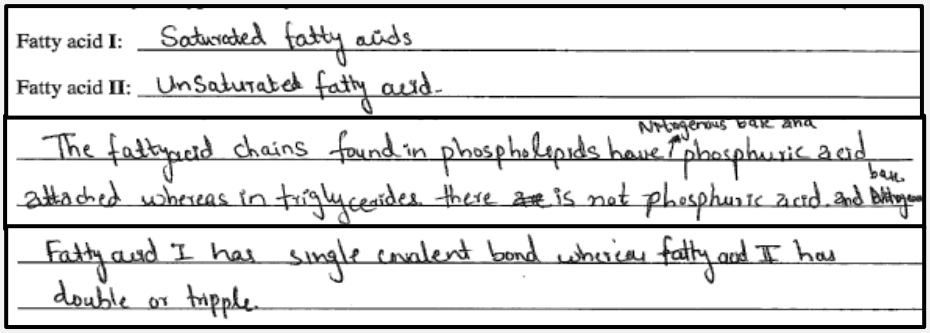
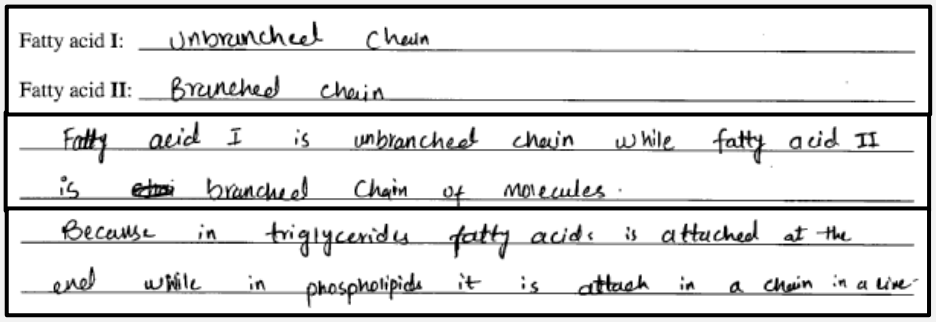
Question No. 1	
Question Text	Consider the given molecular model of a phospholipid molecule.  a. Identify the types of fatty acids, **I** and **II**. b. Write a structural difference between fatty acids **I** and **II**. c. How do the fatty acid chains found in triglycerides differ from those of phospholipids?
SLO No.	2.5.3
SLO Text	Illustrate the molecular structure of an acylglycerol (triglyceride), a phospholipid and a terpene.
Max Marks	3
Cognitive Level	U*
Checking Hints	a. 1 mark for each correct identification (TWO required) b. 1 mark for writing the difference between fatty acids **I** and **II** c. 1 mark for writing the difference between fatty acids of triglycerides and phospholipids (any ONE required)
Overall Performance	The overall performance of candidates in this question was below expectation, indicating a limited understanding of the structure and composition of lipids, particularly phospholipids and triglycerides. Many candidates struggled to accurately identify molecular models and differentiate between key structural features.
Description of Better Responses	*Better*-responses demonstrated a sound understanding of lipid structure. In part 'a', they correctly identified fatty acid **I** as saturated and fatty acid **II** as unsaturated by recognising the presence or absence of double bonds. In part 'b', they clearly described that double bonds in fatty acid **II** cause a kink in the chain, whereas fatty acid **I** remains straight. In part 'c', they accurately distinguished between triglycerides and phospholipids, noting that triglycerides have three fatty acid chains, while phospholipids contain two fatty acids and a phosphate group replacing the third chain on the glycerol backbone.

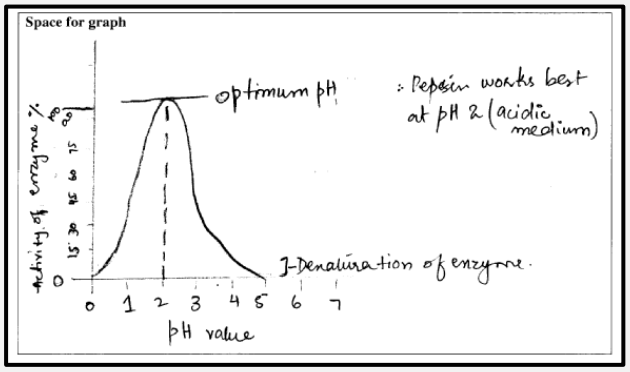
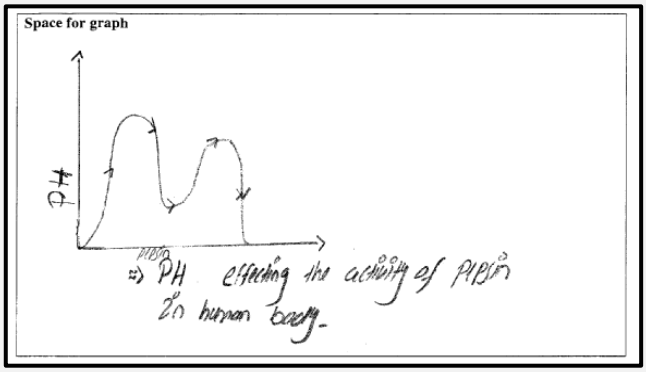
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed poor understanding of lipid structure. In part 'a', many candidates incorrectly identified saturated as unsaturated fatty acids and vice versa. In part 'b', several inaccurately described the difference as branched vs. unbranched chains, while others simply stated that fatty acid **I** is straight and fatty acid **II** is bent, without explaining the role of double bonds. In part 'c', most were unable to differentiate between the fatty acid chains in triglycerides and phospholipids, often omitting the presence of the phosphate group. Responses lacked scientific accuracy and terminology, reflecting weak interpretation of structural models.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Encouraged candidates to use precise scientific terminology, such as saturated, unsaturated, double bonds, phosphate group, and hydrocarbon chain, when describing molecular structures.

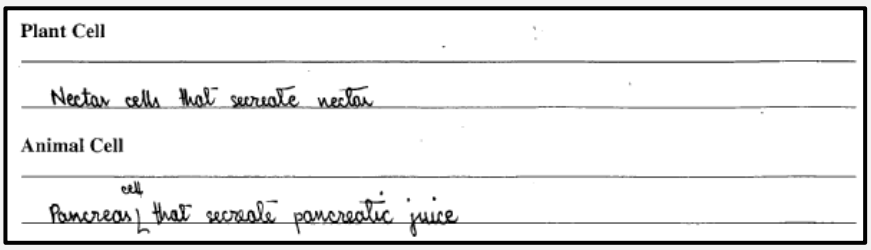
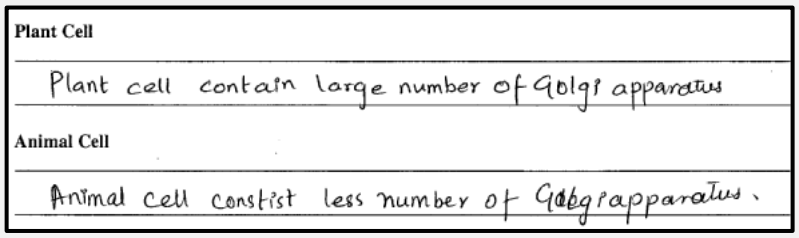
*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 02	
Question Text	Construct a graph in the given space to show the effect of pH on the activity of pepsin in the human digestive system.
SLO No.	3.4.1
SLO Text	Verify the effect of different factors, i.e. pH and temperature on the rate of enzyme action using graph.
Max Marks	2
Cognitive Level	A*
Checking Hints	1 mark for each labelling the x-axis and y-axis 1 mark for depicting the bell shape at acidic pH range
Overall Performance	Most candidates demonstrated a clear understanding of enzyme activity in relation to pH by accurately constructing well-labelled graphs. Overall, the responses reflected strong grasp of enzyme function and appropriate graphing skills. It is highly recommended that when students attempt questions where they have to construct a graph, they should use a darker pencil or pointer pens for clarity of the image and labelling.
Description of Better Responses	*Better responses* demonstrated commendable performance by the candidates on this question. The majority correctly identified that pH should be plotted on the x-axis and enzyme activity on the y-axis of the graph. Additionally, they showed awareness of the optimum pH of pepsin (1.5 to 2.0) in the human digestive system. Candidates accurately placed pH values along the x-axis and enzyme activity along the y-axis. The resulting graphs typically featured a smooth, bell-shaped curve peaking between pH 1.5 and 2.0, with activity decreasing on either side. This clearly illustrates that pepsin is most active in acidic conditions.
Image of Better Responses	
Description of Weaker Responses	*Weaker responses* reflected poor understanding of the effect of pH on enzyme activity and lacked basic graphing accuracy. Several candidates incorrectly labelled the x-axis as temperature instead of pH, showing confusion between different factors affecting enzyme activity. Others mislabelled or omitted axis labels entirely, while some reversed the axes. Many graphs showed incorrect patterns, such as linear trends or multiple peaks, rather than the expected bell-shaped curve with optimal activity at pH 2. A few candidates drew random or flat lines without Scientific justification.
Image of Weaker Responses	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion: Provide regular opportunities for candidates to practice constructing graphs with correctly labelled axes, appropriate scales, and smooth curves. Include tasks that ask candidates to interpret and annotate their graphs.		

Question No. 3	
Question Text	Give ONE example each of a plant cell and an animal cell which contain a large number of Golgi apparatus.
SLO No.	4.4.1
SLO Text	Explain structure, chemical composition and functions of the cellular organelles of animal and plant cell as revealed through the electron microscope: a. cell wall b. cell membrane with reference to Fluid Mosaic Model c. cytoplasm d. endoplasmic reticulum e. ribosomes f. mitochondria g. Golgi apparatus h. lysosomes i. vacuoles j. cytoskeleton k. centrioles l. plastids m. nucleus.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for ONE example of plant cell 1 mark for ONE example of animal cell *(Marks will be awarded for any relevant example)*
Overall Performance	The overall performance of candidates in this question was moderate, as many struggled to connect the function of the Golgi apparatus with specific, real-life examples of cells. This reflects a weakness in applying theoretical knowledge to practical biological contexts.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the function of the Golgi apparatus and its role in secretion. Candidates accurately identified animal cells such as goblet cells or pancreatic cells, as well as plant cells like nectary cells or seed coat cells, both of which require extensive packaging and transport of proteins or other substances.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed considerable gaps in comprehension, indicating a fundamental lack of conceptual understanding of the topic. Several candidates provided vague or incorrect examples, such as 'leaf cell' or 'blood cell', without linking them to secretory functions. Some failed to distinguish between plant and animal cells, while others provided only one example instead of both. A few responses showed confusion about the function of the Golgi apparatus, citing unrelated cell types. Additionally, scientific terminology was often lacking or misused.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

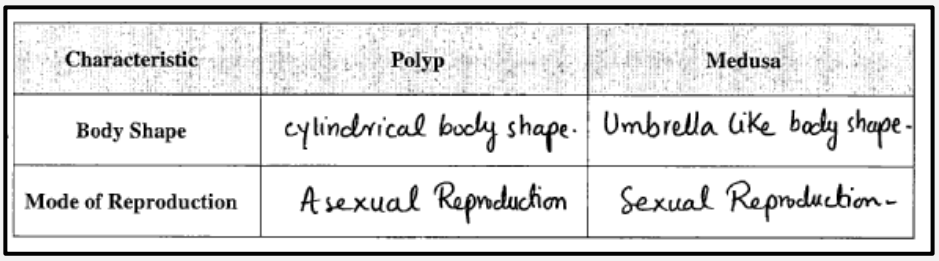
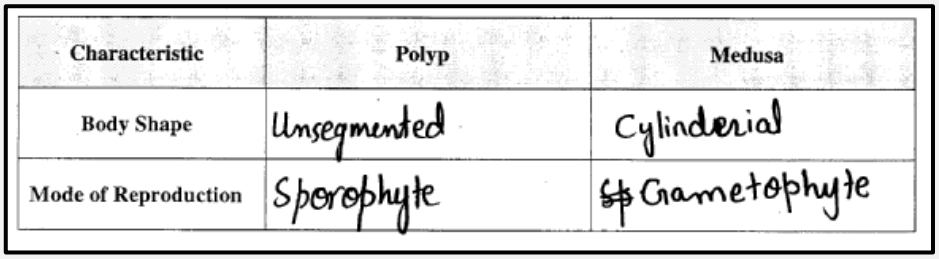
Any Additional Suggestion: Provide candidates with real-life examples of cells rich in different organelles. e.g., cells rich in Golgi apparatus are goblet cells, pancreatic cells, nectary cells, and seed coat cells. Reinforce this understanding through targeted practice questions that require candidates to identify such cells and justify the abundance of multiple organelles based on their functions.

Question No. 4

Question Text	Differentiate between the polyp and medusa of coelenterates in the given table.

Characteristic	Polyp	Medusa
Body Shape		
Mode of Reproduction		

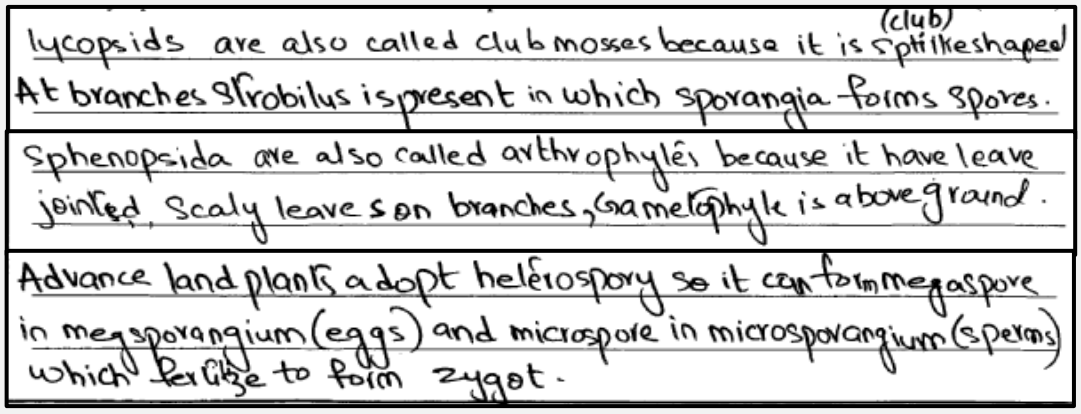
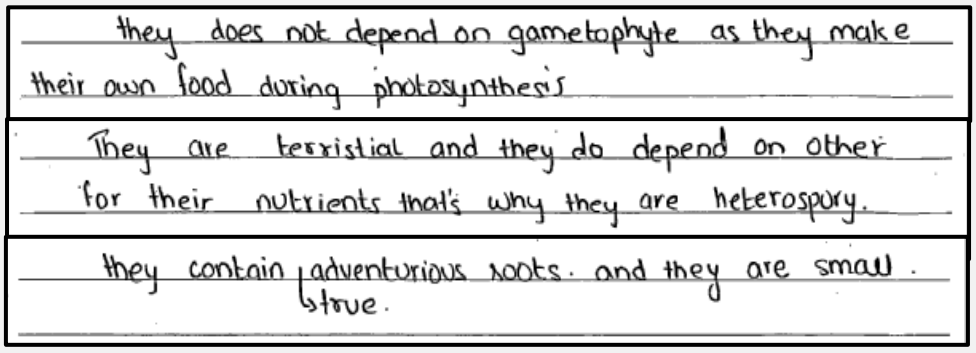
SLO No.	10.4.1

SLO Text	Explain coelenterates with reference to their a. general characteristics b. origin of diploblastic organisation c. radial symmetry d. polymorphism and alternation of generation e. formation of coral reefs.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for each difference (TWO required)
Overall Performance	Candidates generally showed a moderate understanding of the differences between the polyp and medusa forms of coelenterates. Some responses were well-structured, used correct biological terminology, and clearly presented both points of comparison. This reflects a good grasp of coelenterate life cycles and an ability to distinguish between structural and reproductive characteristics.
Description of Better Responses	*Better responses* accurately identified the descriptions of body shape, recognising the polyp as cylindrical or stalked, and the medusa as umbrella-like or saucer-shaped. Reproductive modes were clearly differentiated, with asexual reproduction (budding) associated with polyp form and sexual reproduction (gamete production) with the medusa. Scientific terminology was appropriately and consistently used.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* lacked clarity and demonstrated limited understanding of the distinct body forms. Many candidates gave vague descriptions such as 'polyp is straight' or 'medusa is round,' lacking scientific accuracy. Some confused the reproductive modes or stated only one, often inaccurately stating that both forms reproduce sexually. This reflects an incomplete understanding and a poor grasp of the underlying concepts.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Encourage candidates to create and revise comparison tables highlighting key differences between polyp and medusa forms in terms of shape, mobility, and reproduction. This helps in reinforcing contrasts clearly. Additionally, guide candidates to study labelled diagrams of both forms, as visual learning aids in understanding structural differences more effectively.		

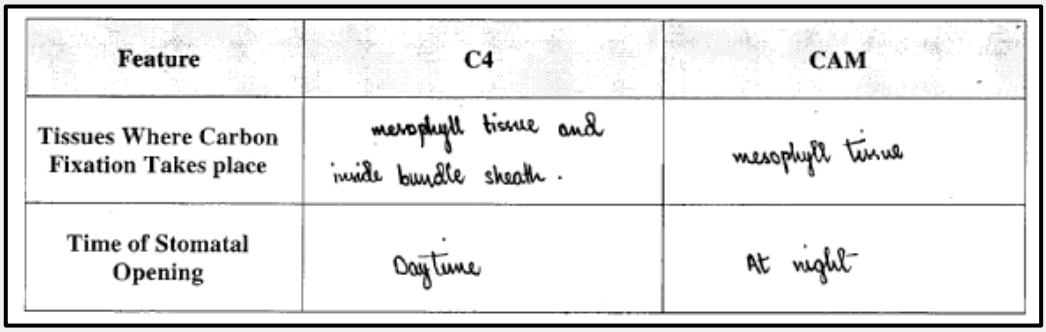
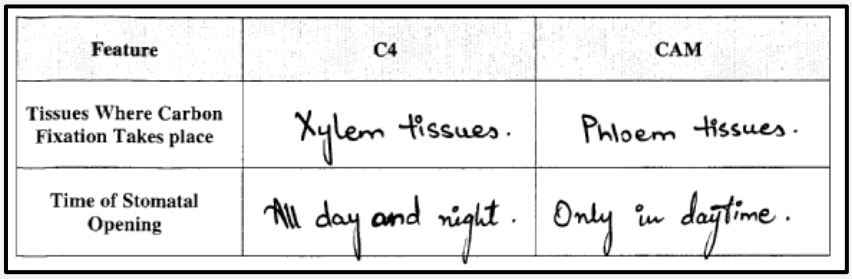
Question No. 5	
Question Text	Give a reason for the following facts: a. Lycopsids are also called club mosses/ spike mosses. b. Sphenopsids are also called arthrophytes. c. Advance land plants adopt heterospory.
SLO No.	9.3.1
SLO Text	Compare major groups of tracheophyta, i.e., a. psilopsida b. lycopsida c. sphenopsida d. Pteropsida.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for each reason (3 required)
Overall Performance	The overall performance of candidates in this question was below average. While most candidates were able to use some relevant biological terms, many struggled to provide accurate and complete reasoning. Overall, responses lacked clarity, detail, and proper biological reasoning.
Description of Better Responses	*Better responses* showed clear understanding and accurate reasoning. In part 'a', candidates correctly stated that lycopsids are referred to as club or spike mosses due to their moss-like appearance and club-shaped strobili. In part 'b', they appropriately linked the term 'arthrophytes' to the jointed stems characteristic of sphenopsids. In part 'c', candidates clearly described that heterospory in advanced land plants enables the production of separate male and female gametophytes or microspores and megaspores, enhancing reproductive success on land. Responses were concise, well-structured, and demonstrated appropriate use of biological terminology.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* reflected limited understanding and revealed several common misconceptions. In part 'a', many candidates confused lycopsids with true mosses, failing to relate the terms 'club' or 'spike' with the shape of their reproductive structures. In part 'b', some incorrectly associated the term 'arthrophytes' with arthropods, rather than recognising its reference to the jointed stem structure of sphenopsids. In part 'c', several candidates misunderstood heterospory as merely the production of different types of spores, without mentioning their distinct reproductive roles or significance. Explanations were often vague, lacked biological reasoning, and showed poor use of terminology, indicating weak conceptual clarity across all three parts.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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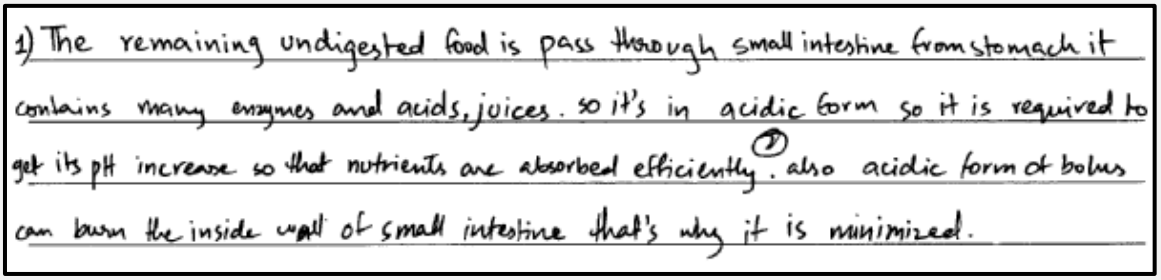
Any Additional Suggestion: Encourage concept-based reasoning by asking candidates to justify scientific names and adaptations using logical, evidence-based explanations.

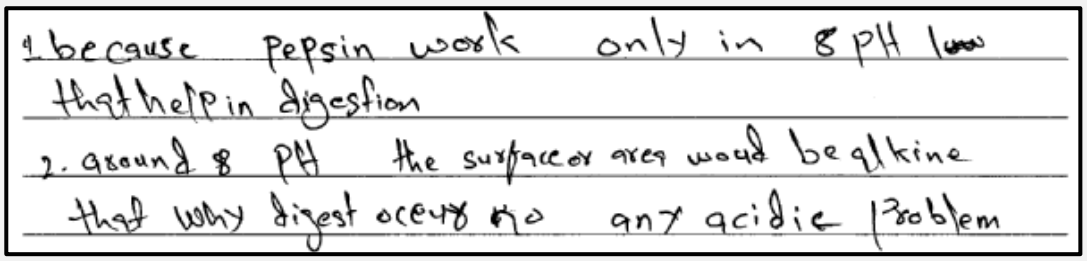
<table>
<tr><th colspan="2">Question No. 6</th></tr>
<tr><td>Question Text</td><td>Differentiate between C4 plants and CAM plants in the given table.
<table>
<tr><th>Feature</th><th>C4</th><th>CAM</th></tr>
<tr><td>Tissues Where Carbon Fixation Takes place</td><td></td><td></td></tr>
<tr><td>Time of Stomatal Opening</td><td></td><td></td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>11.2.9</td></tr>
<tr><td>SLO Text</td><td>Compare C3, C4 and CAM plants.</td></tr>
<tr><td>Max Marks</td><td>2</td></tr>
<tr><td>Cognitive Level</td><td>U</td></tr>
<tr><td>Checking Hints</td><td>1 mark for each difference (TWO required)</td></tr>
<tr><td>Overall Performance</td><td>The overall performance of candidates in this question was average. While many were able to recall the basic differences between C4 and CAM plants, their responses often lacked precision and completeness.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses accurately differentiated between C4 and CAM plants. Candidates correctly explained that C4 plants fix carbon in both mesophyll and bundle sheath cells (spatial separation), with stomata open during the day, whereas CAM plants fix carbon in mesophyll cells only (temporal separation), with stomata open at night. Responses were concise, used correct scientific terminology, and clearly linked structural and functional adaptations to environmental conditions.</td></tr>
<tr><td>Image of Better Response</td><td> 
<table>
<tr><th>Feature</th><th>C4</th><th>CAM</th></tr>
<tr><td>Tissues Where Carbon Fixation Takes place</td><td>mesophyll tissue and inside bundle sheath.</td><td>mesophyll tissue</td></tr>
<tr><td>Time of Stomatal Opening</td><td>Daytime</td><td>At night</td></tr>
</table></td></tr>
<tr><td>Description of Weaker Responses</td><td>Weaker responses showed limited understanding of the differences between C4 and CAM plants. Many candidates struggled to correctly identify the tissues involved in carbon fixation, often failing to mention the bundle sheath cells in C4 plants or incorrectly stating that both types fix carbon in the same tissues without distinction. Additionally, several candidates incorrectly reversed the stomatal opening times, stating that C4 plants open their stomata at night and CAM plants during the day. Overall, these responses lacked clarity, correct terminology, and accurate application of key concepts.</td></tr>
<tr><td>Image of Weaker Response</td><td> 
<table>
<tr><th>Feature</th><th>C4</th><th>CAM</th></tr>
<tr><td>Tissues Where Carbon Fixation Takes place</td><td>Xylem tissues.</td><td>Phloem tissues.</td></tr>
<tr><td>Time of Stomatal Opening</td><td>All day and night.</td><td>Only in daytime.</td></tr>
</table></td></tr>
</table>

Suggestions for improvement (Highlight all that apply)

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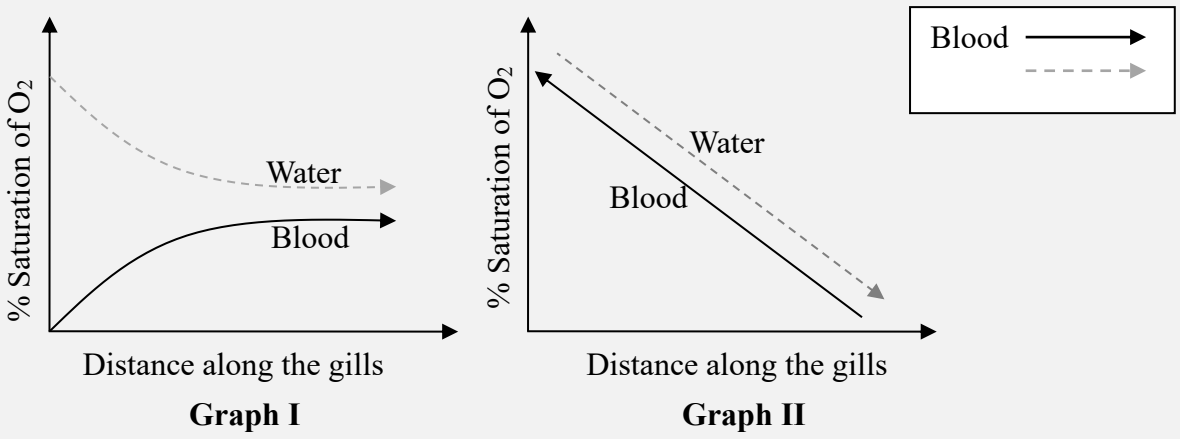
Any Additional Suggestion: Use contextual learning by relating each photosynthetic pathway to specific environmental adaptations, e.g., associating C_4 plants with hot, sunny environments and CAM plants with arid, dry regions, to help candidates understand their ecological significance.

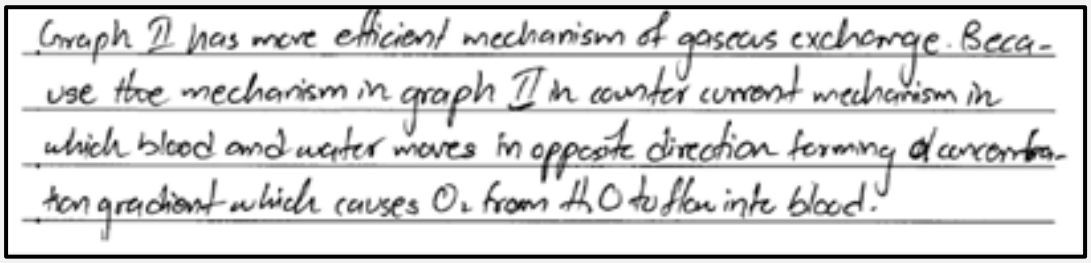
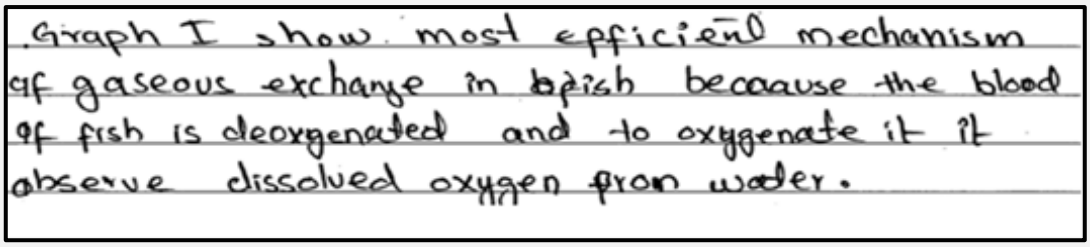
Question No. 7	
Question Text	Why does an alkaline pH (around 8) is necessary for the human small intestine to function efficiently? Give TWO reasons.
SLO No.	12.4.2
SLO Text	Discuss the process of digestion of carbohydrates, proteins and lipids in human being.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for each reason (any TWO required)
Overall Performance	The overall performance on this question was moderate. A number of candidates demonstrated confusion between enzymes and hormones, indicating a lack of clarity in foundational biological concepts.
Description of Better Responses	*Better responses* reflected a clear and accurate understanding of the importance of maintaining an alkaline pH (approximately 8) for the optimal functioning of the human small intestine. Candidates effectively described the role of this pH in facilitating the activity of trypsin, protecting the intestinal epithelium, and promoting efficient nutrient absorption.
Image of Better Response	

Description of Weaker Responses	*Weaker responses* were marked by a lack of clarity and demonstrated a limited understanding of the topic. Many candidates were unaware of the specific pH requirements for enzymes such as pepsin and trypsin, and their responses revealed several misconceptions. Responses often incorrectly linked the question to unrelated issues such as surface area or general acidity, rather than addressing the biochemical basis of enzyme function.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Strengthen candidates' understanding of pH-specific enzyme action by integrating theoretical explanations with hands-on demonstrations, laboratory experiments, or digital simulations that illustrate enzyme activity under varying pH conditions.		

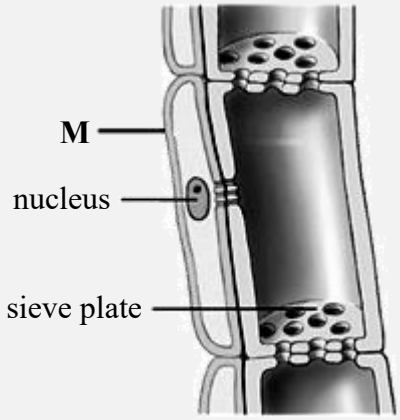
Question No. 8	
Question Text	The given graphs (**I** and **II**) show two mechanisms of gaseous exchange in fish.  **Graph I** **Graph II**

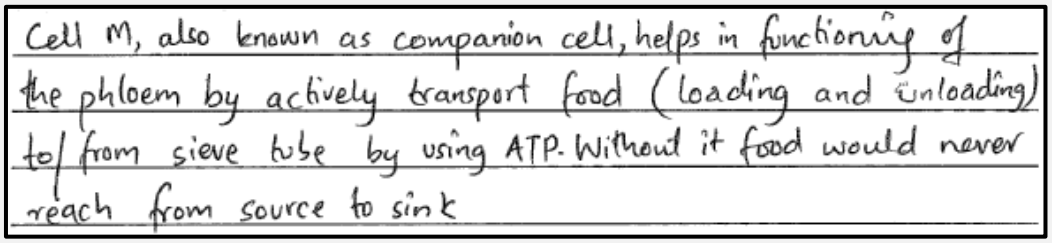
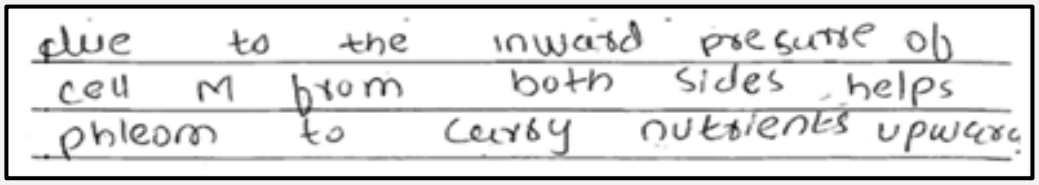
	Which graph (**I or II**) represents the most efficient mechanism of gaseous exchange? Support your answer with a valid reason
SLO No.	13.2.3
SLO Text	Explain the mechanism of gaseous exchange in fish, frogs and birds.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for identification of graph **II** 1 mark for a valid reason (underlined)
Overall Performance	The overall performance on this question was slightly above average. Some candidates identified the graph wrongly, reflecting a general lack of clarity and conceptual depth. This indicates the need to strengthen candidates' understanding of physiological mechanisms and enhance their ability to interpret graphs and extract relevant information effectively.
Description of Better Responses	*Better responses* correctly identified Graph **II** as representing the most efficient mechanism of gaseous exchange. These candidates accurately explained that it illustrates the counter-current exchange system, which maximises oxygen absorption by maintaining a concentration gradient along the entire length of the gill surface, allowing continuous diffusion of oxygen into the blood.
Image of Better Responses	
Description of Weaker Responses	*Weaker responses* reflected significant misconceptions and a lack of understanding of the underlying principles of gaseous exchange in fish. Many candidates incorrectly identified the less efficient mechanism (parallel flow, graph **I**) as the better option, often basing their responses on assumptions unrelated to the graph's data. A common misconception was the belief that any overlap between oxygen concentrations indicates greater efficiency. Others confused the concept of diffusion or failed to mention the importance of maintaining a concentration gradient. Additionally, several responses showed difficulty in interpreting the graphs, indicating weak data analysis skills and limited familiarity with key terms such as counter current flow, diffusion gradient, and oxygen uptake.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Incorporate regular practice of interpreting biological graphs and data. Guide candidates' on how to extract trends, compare curves, and relate graphical information to underlying biological processes.

Question No. 9	
Question Text	The given diagram shows some specialised cells of a plant.  Briefly describe how cell **M** is necessary for the functioning of the phloem.
SLO No.	14.5.2
SLO Text	Explain mechanism of phloem translocation, i.e. diffusion and pressure flow hypothesis.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for identification of cell **M** 1 mark for describing any ONE function
Overall Performance	The overall performance of candidates was average, reflecting limited understanding of the interdependent relationship between companion cells and sieve tube elements in phloem transport. Many responses lacked clarity on how companion cells support the metabolic

	activities and loading/ unloading processes essential for effective translocation. This suggests a need for more targeted practice with diagram-based questions to strengthen candidates' visual interpretation skills and application of biological concepts.
Description of Better Responses	*Better responses* demonstrated a sound understanding of the role of companion cells in phloem function. These candidates correctly identified cell **M** as a companion cell and clearly explained its role in providing metabolic support to sieve tube elements. They accurately mentioned processes such as the active loading and unloading of sugars, as well as the maintenance of the pressure gradient necessary for translocation. In addition, these responses reflected strong diagram interpretation skills and precise application of relevant biological terminology.
Image of Better Responses	 Cell M, also known as companion cell, helps in functioning of the phloem by actively transport food (loading and unloading) to/ from sieve tube by using ATP. Without it food would never reach from source to sink
Description of Weaker Responses	*Weaker responses* reflected confusion about the identity and function of cell **M**. Many candidates failed to recognise it as a companion cell, with some incorrectly identifying it as a xylem component or a general plant cell. Several responses were vague or inaccurate, lacking references to essential functions such as active transport of sugars, metabolic support, or the role in maintaining the pressure gradient in phloem translocation. These misconceptions, along with the inability to extract information from the diagram, indicate a weak conceptual grasp and highlight the need for targeted practice with diagram-based questions and the roles of specialised cells in transport systems.
Image of Weaker Response	 due to the inward pressure of cell M from both sides helps phloem to carry nutrients upwards

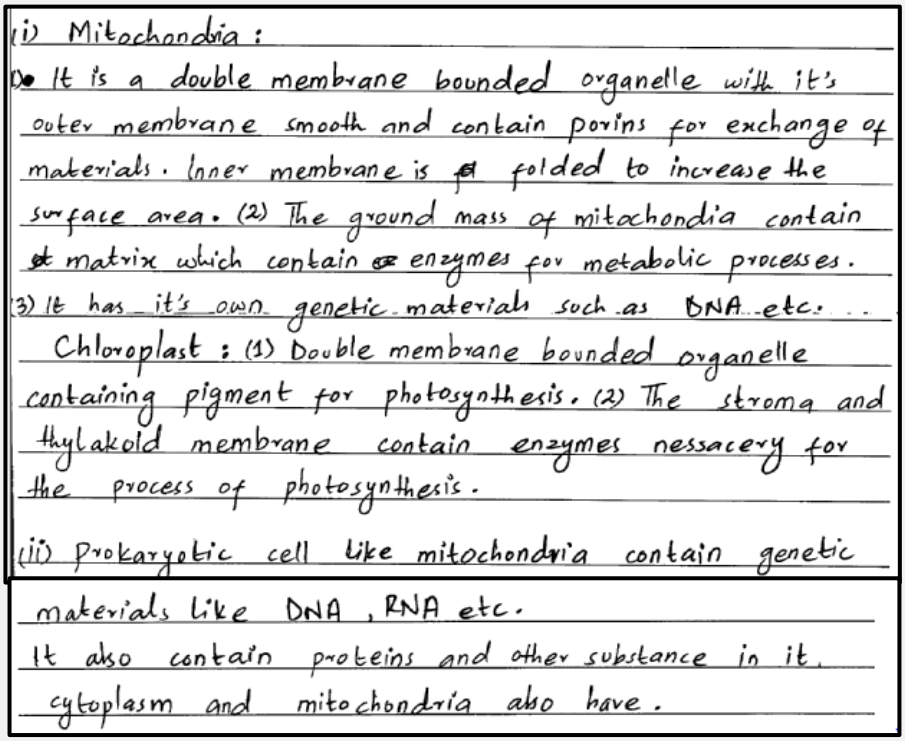
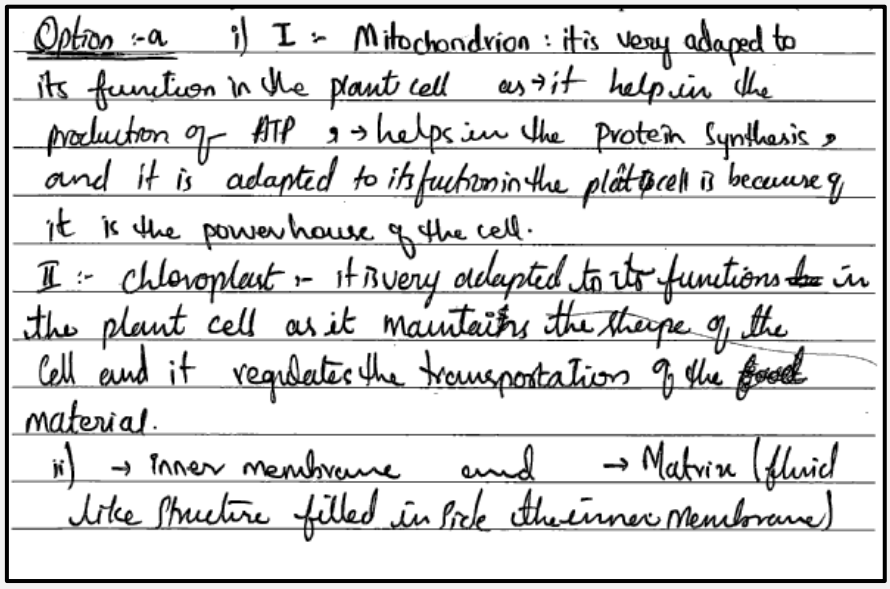
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion: Regularly incorporate visual-based questions in assessments and classroom activities to build candidates' skills in interpreting and responding to diagrammatic information.		

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts ‘a’ and ‘b’

Question No. 10(a)	
Question Text	i. Explain how the following cell organelles are adapted to their functions in a plant cell. I. Mitochondrion (any THREE points) II. Chloroplast (any TWO points) ii. Identify any TWO subcellular structures in a prokaryotic cell that resemble to those found in a mitochondrion.
SLO No.	4.4.1
SLO Text	Explain structure, chemical composition and functions of the cellular organelles of animal and plant cell as revealed through the electron microscope: a. cell wall b. cell membrane with reference to Fluid Mosaic Model c. cytoplasm d. endoplasmic reticulum e. ribosomes f. mitochondria g. Golgi apparatus h. lysosomes i. vacuoles j. cytoskeleton k. centrioles l. plastids m. nucleus.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for explaining each adaptation of mitochondrion (any THREE required) 1 mark for explaining each adaptation of chloroplast (any TWO required) ii. 1 mark for identifying each similar structure between mitochondrion and prokaryotic cell (any TWO required)
Overall Performance	Overall, part ‘a’ was attempted by most candidates as compared to part ‘b’. Performance in this section was generally average. Many responses lacked appropriate scientific terminology and did not reflect the depth of understanding expected at the Grade 11 level by not mentioning the structural adaptations of mitochondria and chloroplast. This indicates the need for greater emphasis on conceptual clarity and the application of precise biological language in written responses
Description of Better Responses	*Better responses* showed a clear understanding of how structural features support the functions of mitochondria and chloroplasts. Candidates accurately mentioned cristae for increased surface area, enzymes for respiration, and the presence of double membranes in mitochondria. They also identified thylakoids and stroma in chloroplasts as essential for photosynthesis. In part (ii), well-prepared candidates correctly identified ribosomes and circular DNA in prokaryotes as features shared with mitochondria, showing the ability to draw accurate comparisons using appropriate scientific terminology. These responses reflected strong content knowledge and the ability to apply it concisely.

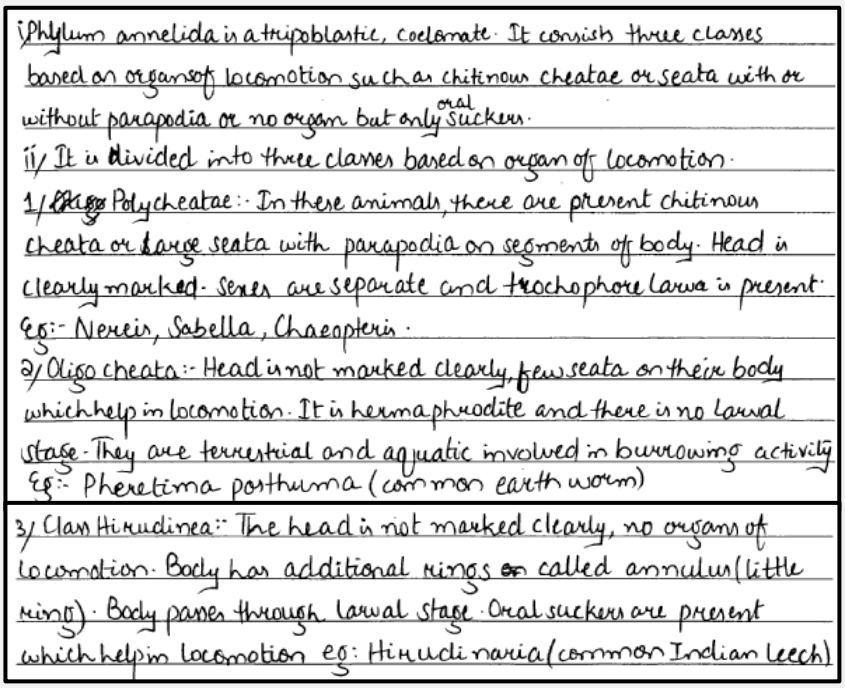
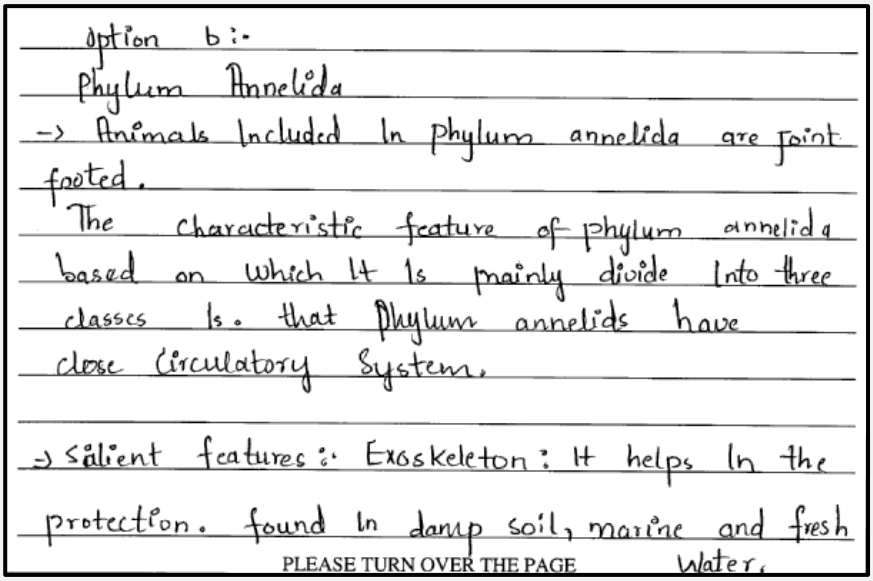
Image of Better Responses	 (i) Mitochondria : • It is a double membrane bounded organelle with it's outer membrane smooth and contain porins for exchange of materials. Inner membrane is folded to increase the surface area. (2) The ground mass of mitochondria contain matrix which contain enzymes for metabolic processes. (3) It has it's own genetic materials such as DNA etc. Chloroplast : (1) Double membrane bounded organelle containing pigment for photosynthesis. (2) The stroma and thylakoid membrane contain enzymes nessacery for the process of photosynthesis. (ii) Prokaryotic cell like mitochondria contain genetic materials like DNA , RNA etc. It also contain proteins and other substance in it. cytoplasm and mitochondria also have.
Description of Weaker Responses	*Weaker responses* reflected limited understanding of the organelle structure-function relationships. Many candidates gave vague or overly general answers, such as 'mitochondria make energy' without referring to specific adaptations like cristae or enzymes. Some confused the roles of mitochondria and chloroplasts or described features irrelevant to plant cells. In part 'ii', common misconceptions included identifying structures like the nucleus or cell wall as similar to mitochondrial components, showing confusion between prokaryotic and eukaryotic cell features. These responses indicate gaps in foundational cell biology and a lack of familiarity with appropriate scientific terminology.
Image of Weaker Response	 Option :-a i) I :- Mitochondrion : it is very adaped to its function in the plant cell as → it help in the production of ATP , → helps in the protein Synthesis , and it is adapted to its function in the plant cell is because of it is the powerhouse of the cell. II :- chloroplast :- it is very adapted to its functions in the plant cell as it maintains the shape of the Cell and it regulates the transportation of the food material. ii) → Inner membrane and → Matrix (fluid like structure filled in side the inner membrane)

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Provide scaffolded support and differentiated extension tasks to address varied ability levels, ensuring all candidates can effectively grasp and apply key biological concepts. Additionally, implement regular formative assessments and provide immediate, targeted feedback to identify and address misconceptions at an early stage.

Question No. 10(b)	
Question Text	i. Identify any ONE characteristic features of phylum annelida based on which it is mainly divided into three different classes? ii. Given are the two species of phylum annelida. • *Pheretima posthuma* (Earthworm) • *Hirudinaria* (Leech) I. Name the classes of phylum annelida to which each of the given species belong. II. Write any TWO salient features of each class named in part 'I'.
SLO No.	10.7.2
SLO Text	Classify annelides up to classes.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for identifying a characteristic feature (any ONE required) ii. I. 1 mark for naming the class to which each species belongs (2 required) II. 1 mark for each salient feature of class to which earthworm belongs (2 required) 1 mark for each salient feature of class to which leech belongs (2 required)
Overall Performance	This part of the question was attempted by a smaller number of candidates; however, those who did attempt it generally performed well. Candidates demonstrated a clear understanding of the classification within the phylum Annelida. Their responses were well-structured, used appropriate biological terminology, and reflected a sound grasp of invertebrate classification principles.

Description of Better Responses	*Better responses* accurately identified key characteristics used in the classification of phylum Annelida, such as the number and type of setae, the presence or absence of parapodia, and the structure of metameres. Candidates correctly classified *Pheretima posthuma* under Oligochaeta and *Hirudinaria* under Hirudinea, clearly stated two relevant features for each class. These responses demonstrated a strong understanding of annelid diversity, appropriate use of scientific terminology, and the ability to connect structural features with taxonomic classification.
Imageof Better Responses	
Description of Weaker Responses	*Weaker responses* reflected a lack of understanding of the basis for classification within phylum Annelida. In part (i), most candidates were unable to identify key distinguishing features such as the number and type of setae, presence of parapodia, and the segmentation of the body (metameres). In part (ii), some incorrectly classified *Pheretima posthuma* (earthworm) as a Polychaete and *Hirudinaria* (leech) as an Oligochaete, indicating confusion between annelid classes. In part (iii), several candidates listed general characteristics of Annelida rather than class-specific features, demonstrating limited knowledge of class-level distinctions and a weak application of biological classification principles.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Encourage candidates to practice and build step-by-step accuracy, guiding candidates' to first identify key classification features, then assign organisms to the correct classes, and finally list class-specific features.		

Question No. 11(a)	
Question Text	i. Identify any TWO major components of tobacco smoke and describe how each of them causes damage to the human respiratory system. ii. Describe the effect of each of the following respiratory diseases on the human respiratory system. I. Chronic Bronchitis II. Emphysema III. Lung cancer
SLO No.	13.4.2
SLO Text	Explain the effects of smoking on respiratory system.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for identifying each component of tobacco smoke (any TWO required) 1 mark for describing each component of tobacco smoke (any TWO required) ii.1 mark for describing each respiratory disease (THREE required)
Overall Performance	Overall, part 'a' was attempted by most candidates as compared to part 'b'. However, overall performance on this question was below expectations. Many candidates demonstrated a lack of knowledge regarding the key components of tobacco smoke, such as tar, carbon monoxide, heavy metals, and nicotine. Additionally, several candidates confused the specified respiratory diseases with unrelated conditions, indicating a fundamental misunderstanding of the topic.

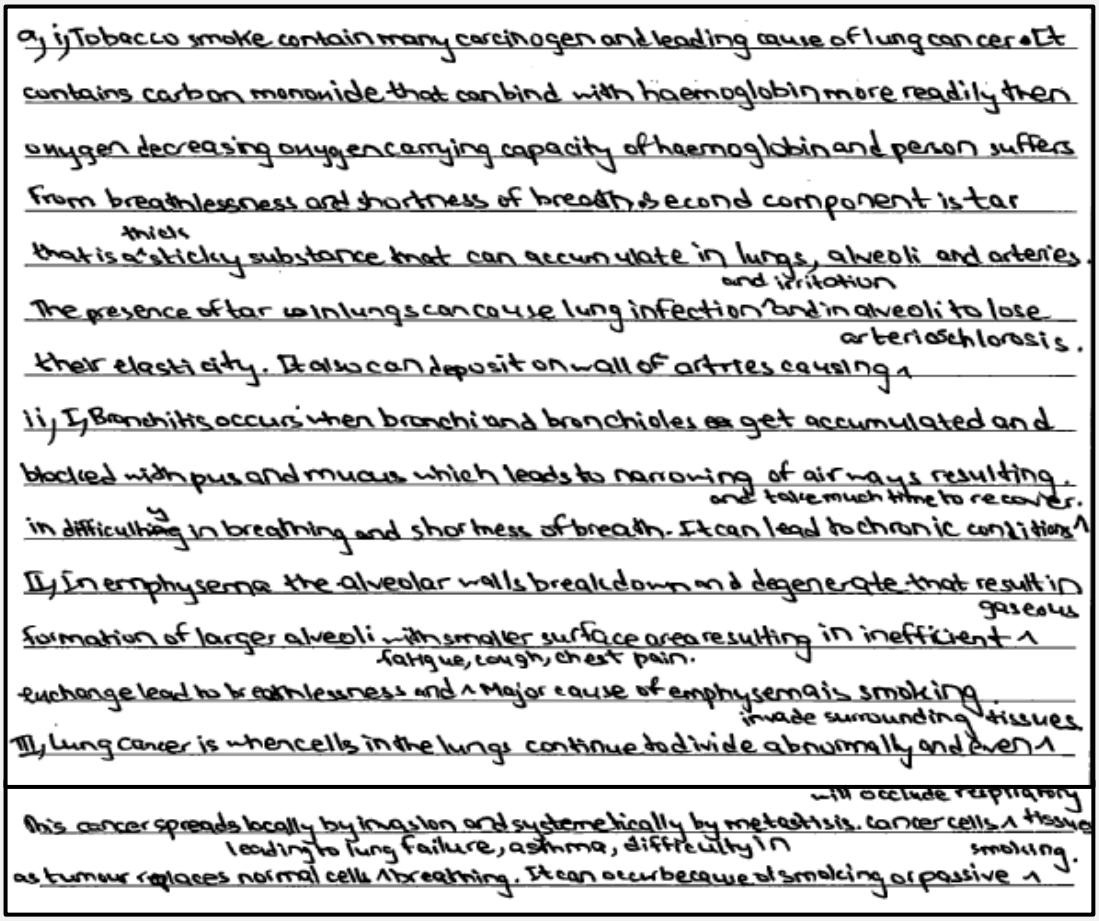
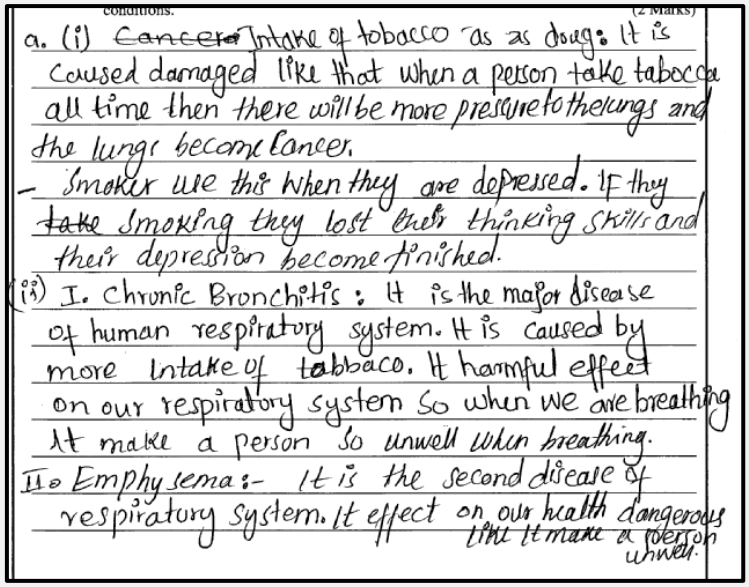
Description of Better Responses	*Better responses* accurately identified harmful components of tobacco smoke in part 'i' such as tar, carbon monoxide, and nicotine. Candidates correctly explained that tar contains carcinogens and stains lung tissue, carbon monoxide reduces oxygen transport by binding to haemoglobin, and nicotine increases mucus production and damages cilia, thereby leading to reduced airway clearance. In part 'ii', candidates effectively described how chronic bronchitis causes excess mucus and damaged cilia, how emphysema reduces the surface area for gas exchange due to alveolar damage, and how lung cancer disrupts airflow and impairs oxygen intake.
Images of Better Responses	
Description of Weaker Responses	*Weaker responses* were often overly generalised and lacked detail regarding the damaging effects of specific components of tobacco smoke. In part 'i', many candidates simply stated that smoking is harmful without explaining how substances like tar, carbon monoxide, or nicotine affect the respiratory system. A common misconception was that nicotine causes cancer or that carbon monoxide directly damages lung tissue, rather than impairing oxygen transport. In part 'ii', some candidates confused the symptoms of respiratory diseases or gave vague descriptions without linking them to the underlying physiological damage. These responses reflected a superficial understanding of the topic and limited use of appropriate scientific terminology.

Image of Weaker Response	

conditions. (2 Marks)

a. (i) ~~Cancer~~ Intake of tobacco as as drug: It is caused damaged like that when a person take tabacco all time then there will be more pressure to the lungs and the lungs become cancer.
- Smoker use this when they are depressed. If they ~~take~~ Smoking they lost their thinking skills and their depression become finished.

(ii) I. Chronic Bronchitis : It is the major disease of human respiratory system. It is caused by more intake of tabbaco. It harmful effect on our respiratory system so when we are breathing it make a person so unwell when breathing.

II. Emphysema :- It is the second disease of respiratory system. It effect on our health dangerous that it make a person unwell.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Provide model answers and guided writing frames by using our resource guide available on AKU-EB website, to help candidates structure their responses with clear cause-and-effect links and scientific precision.

Question No. 11(b)	
Question Text	i. Describe the pathway of cardiac conduction in relation to the mechanism of excitation and contraction of the heart. ii. Write any TWO situations where an artificial pacemaker is employed due to abnormal conditions.
SLO No.	14.7.5 and 14.7.6
SLO Text	Explain cardiac cycle (sequence of events and mechanism of heart excitation and contraction); relate the function of the artificial pacemaker with that of the sino-atrial node.

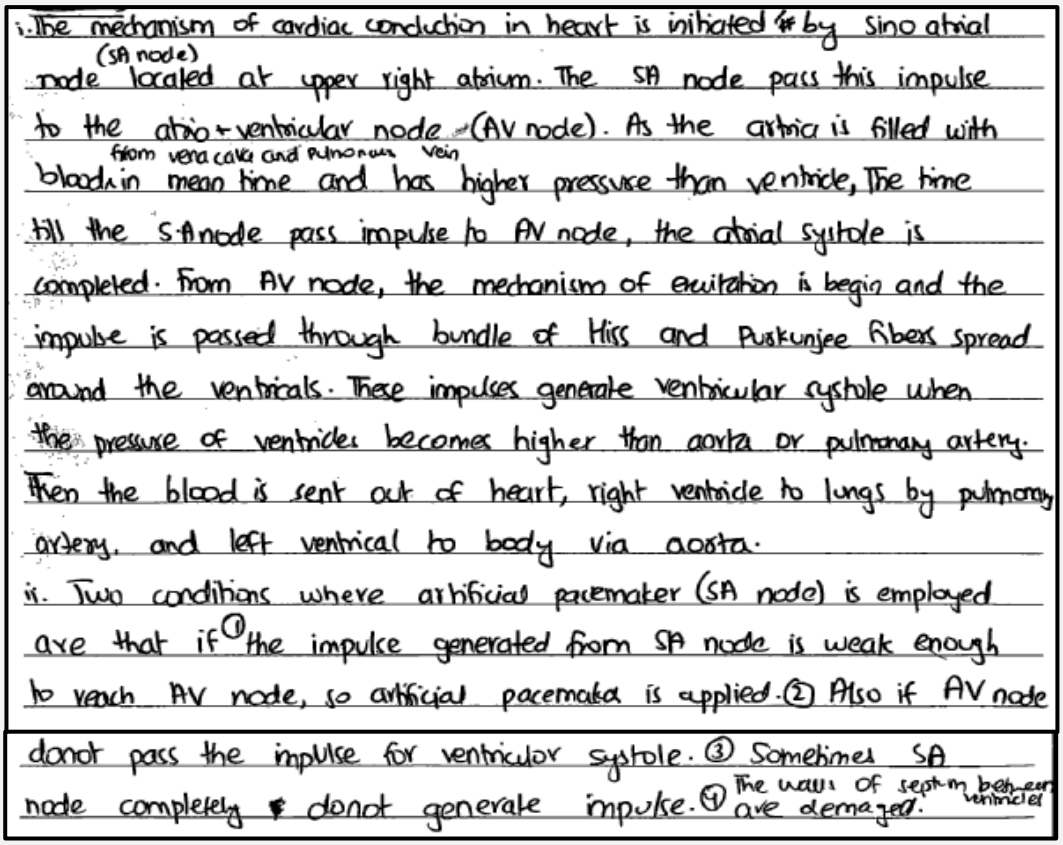
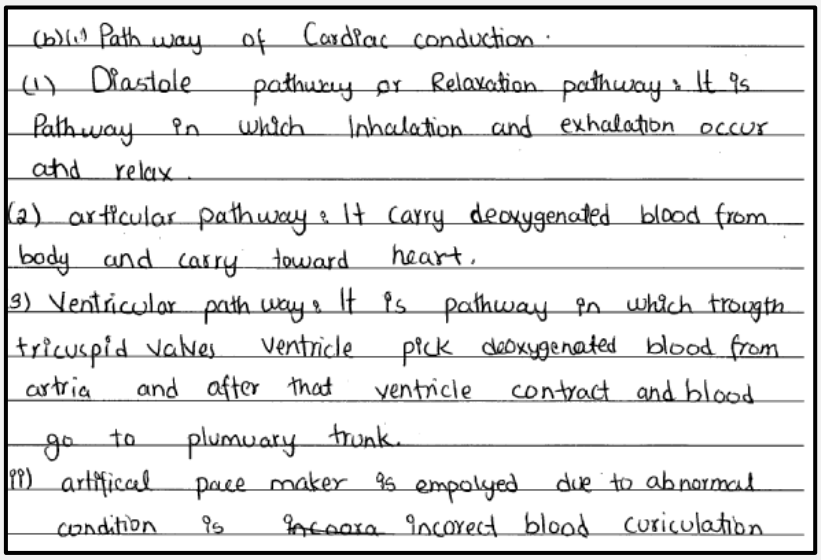
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for each underlined point (FIVE required) ii. 1 mark for each condition (any TWO required)
Overall Performance	The overall performance of candidates on this question was average, reflecting limited clarity and conceptual depth, and indicating a generally superficial understanding of the topic.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the cardiac conduction pathway. In part 'i', accurately describing the roles of the sino-atrial node (SAN), atrioventricular node (AVN), bundle of His, and purkinje fibres, including the crucial delay between atrial and ventricular contractions. Candidates effectively linked electrical impulses to the mechanical contraction of the heart. In part 'ii', they correctly identified relevant clinical conditions requiring artificial pacemakers, such as electrical conduction blocks and bradycardia, providing specific examples. Overall, these responses were well-organised, detailed, and showed strong command of both physiological mechanisms and clinical applications related to cardiac function and pacemaker use.
Image of Better Responses	 i. The mechanism of cardiac conduction in heart is initiated by sino atrial node (SA node) located at upper right atrium. The SA node pass this impulse to the atrio-ventricular node (AV node). As the atria is filled with blood from vena cava and pulmonary vein in mean time and has higher pressure than ventricle, The time till the SA node pass impulse to AV node, the atrial systole is completed. From AV node, the mechanism of excitation is begin and the impulse is passed through bundle of Hiss and Purkunjee fibers spread around the ventricals. These impulses generate ventricular systole when the pressure of ventricles becomes higher than aorta or pulmonary artery. Then the blood is sent out of heart, right ventricle to lungs by pulmonary artery, and left ventrical to body via aorta. ii. Two conditions where artificial pacemaker (SA node) is employed are that if ① the impulse generated from SA node is weak enough to reach AV node, so artificial pacemaker is applied. ② Also if AV node donot pass the impulse for ventricular systole. ③ Sometimes SA node completely donot generate impulse. ④ The walls of septum between ventricles are damaged.
Description of Weaker Responses	*Weaker responses* often confused the cardiac conduction pathway with the cardiac cycle, focusing on mechanical phases like atrial and ventricular systole instead of describing the flow of electrical impulses. In part (i), key components such as the sino-atrial node, atrio-ventricular node, bundle of His and Purkinje fibres were frequently omitted, and the crucial delay at the AV node was rarely mentioned. In part (ii), candidates often gave overgeneralised responses such as 'the heart is not working' or 'not beating', without specifying conditions like malfunctioning of SA node or AV block. These responses reflected limited understanding and a failure to accurately apply physiological concepts.

Image of Weaker Response	

(b)(i) Path way of Cardiac conduction.
(1) Diastole pathway or Relaxation pathway: It is Pathway in which Inhalation and exhalation occur and relax.
(2) arttficular pathway: It carry deoxygenated blood from body and carry toward heart.
3) Ventricular path way: It is pathway in which trougth tricuspid valves ventricle pick deoxygenated blood from artria and after that ventricle contract and blood go to plumuary trunk.
(ii) artifical pace maker is empolyed due to abnormal condition is ~~Increase~~ incorect blood coriculation

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Integrate animations or ECG recordings into lessons to visually demonstrate how electrical impulses regulate the heartbeat and to illustrate where disruptions in the conduction pathway may require the use of artificial pacemakers by using our resource guide available on AKU-EB website.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, candidates are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, candidates study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).
Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, candidates use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing candidates' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where candidates ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Candidates first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the candidates engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Candidates observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

These contributors include:

- Sajida Mohammed Afzal, Specialist Biology, Curriculum and Examination Development, AKU-EB
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